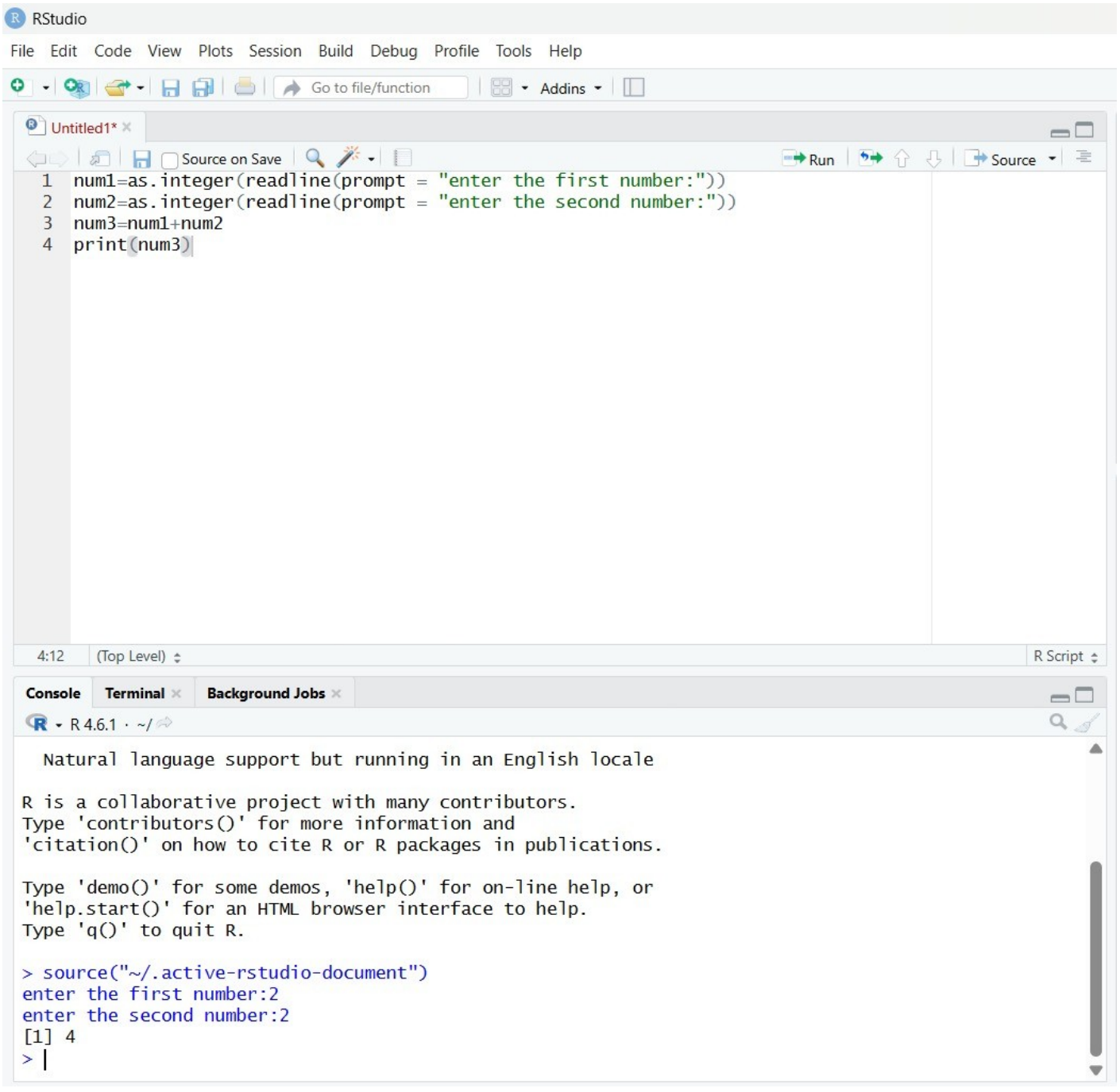
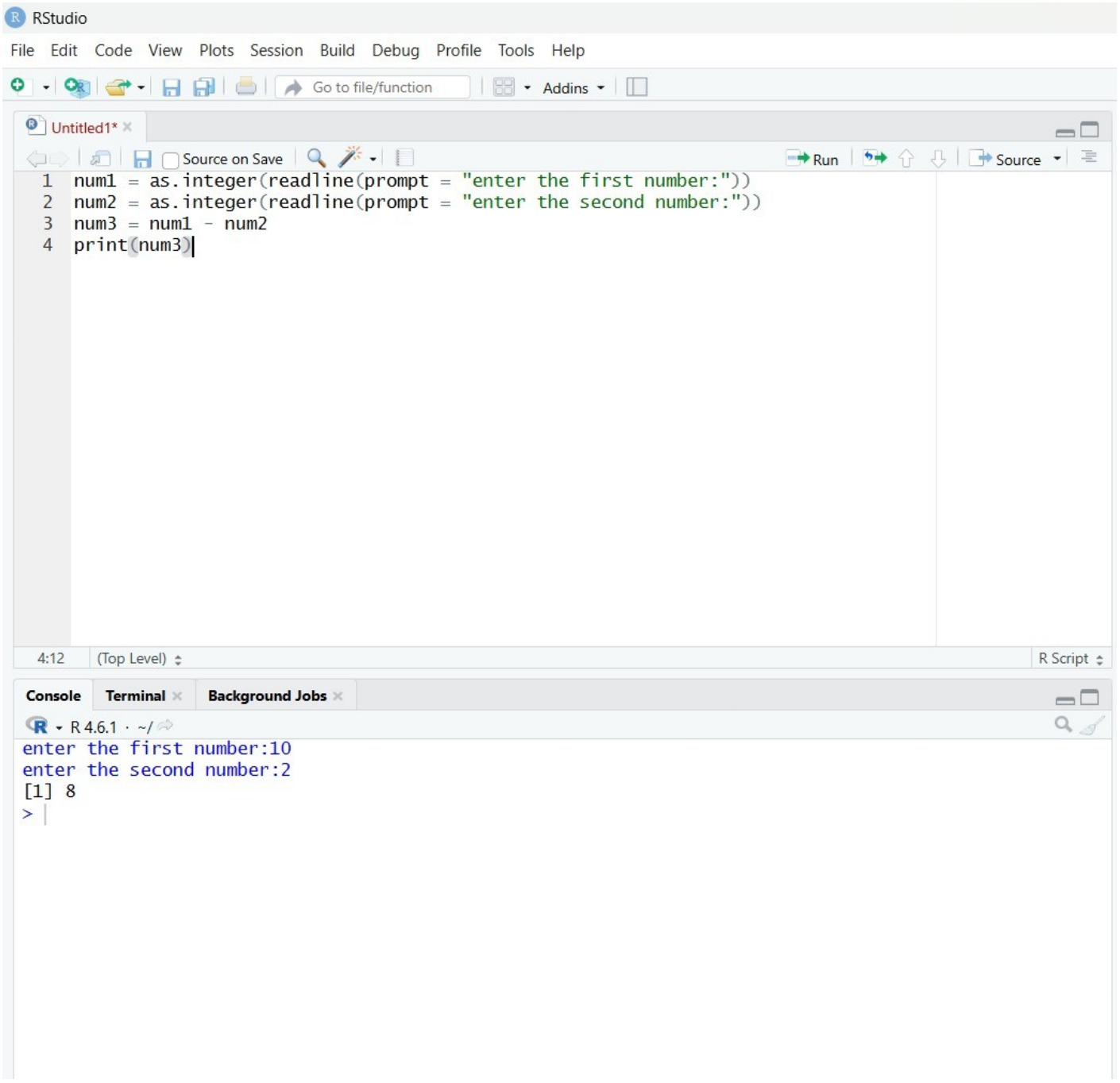
Name: Leo Leenas A
Register Number: 192511068

LAB EXPERIMENTS

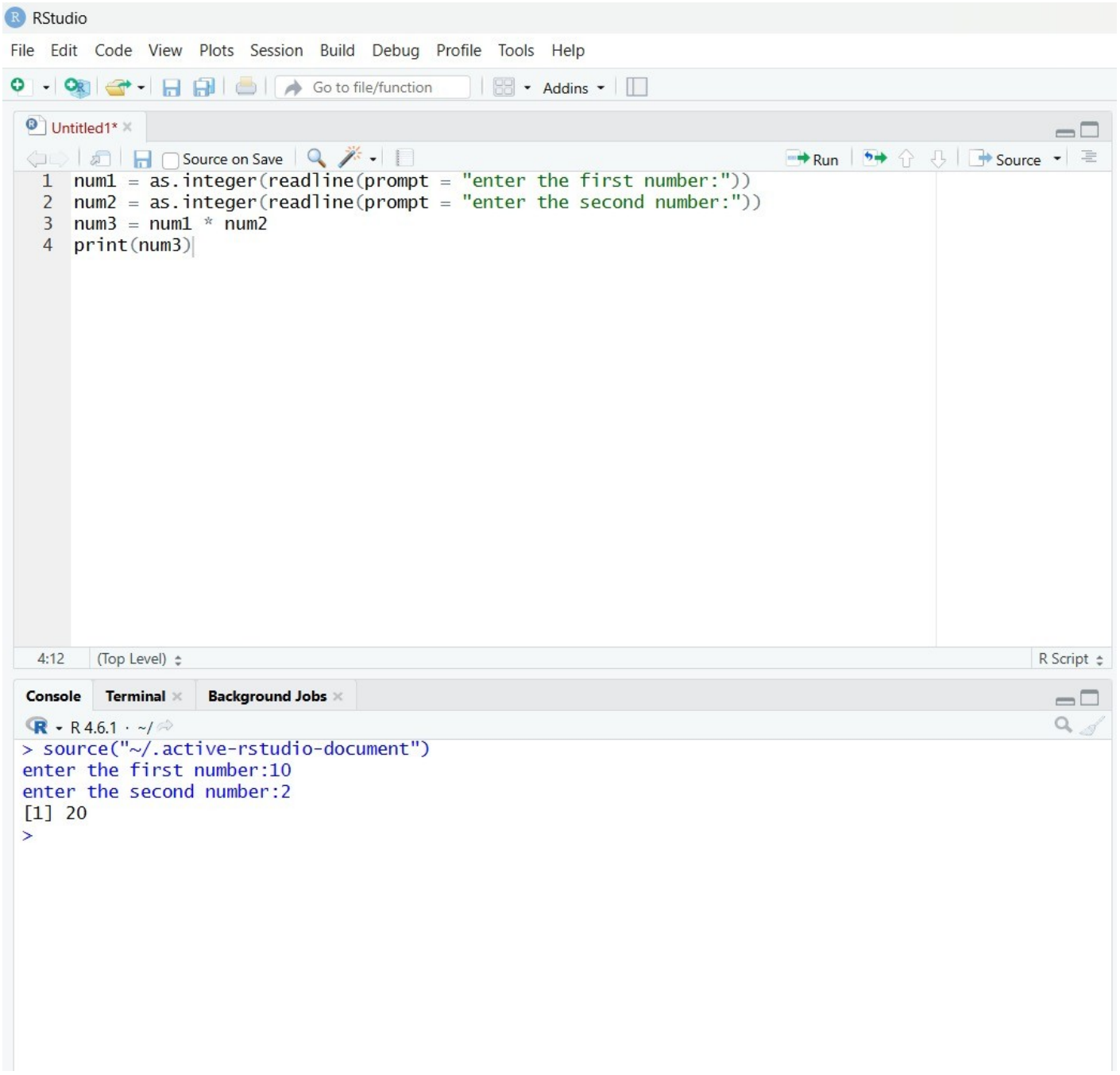
EXPERIMENT-1
ADDITION



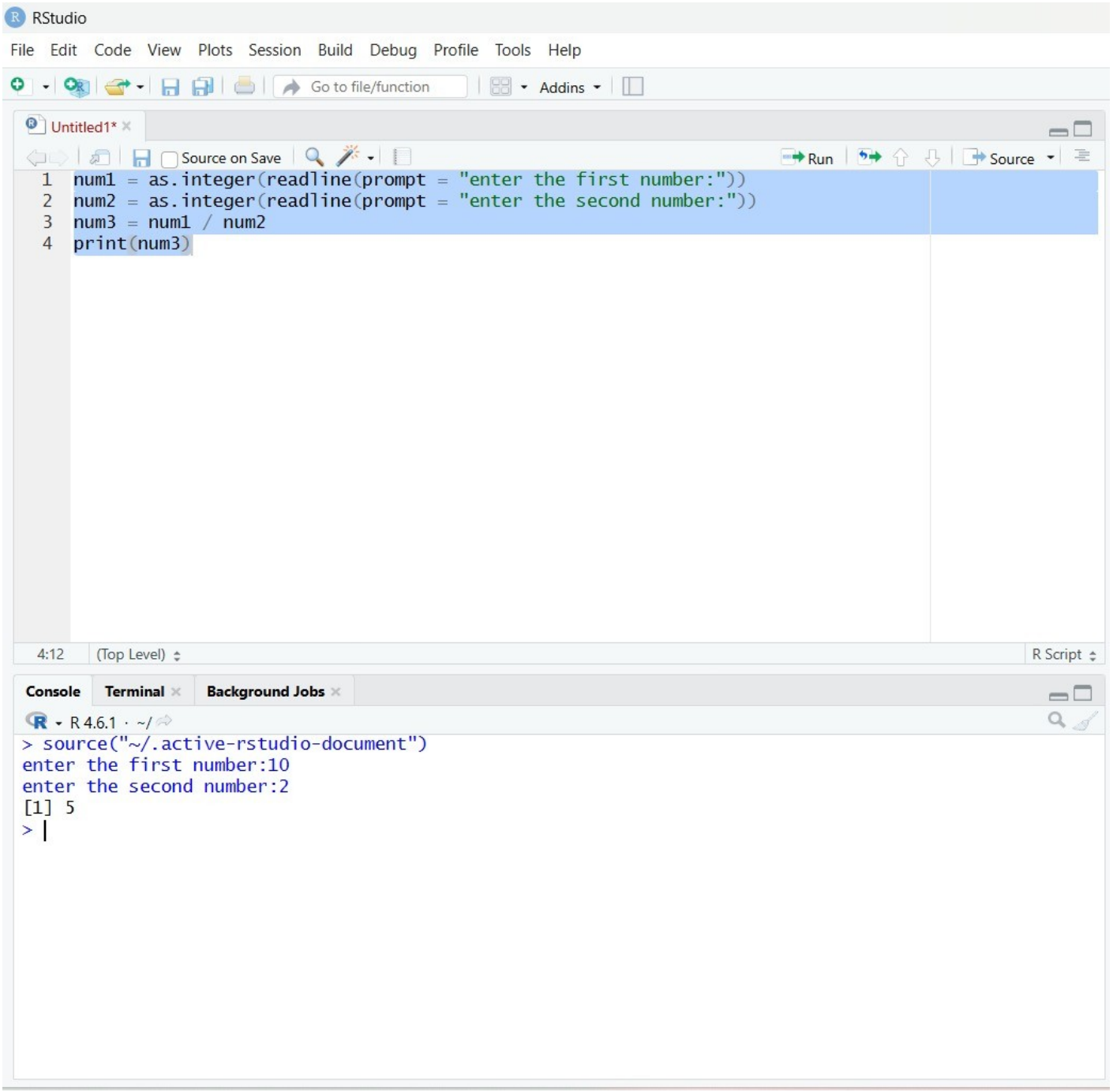
EXPERIMENT-2
SUBTRACTION



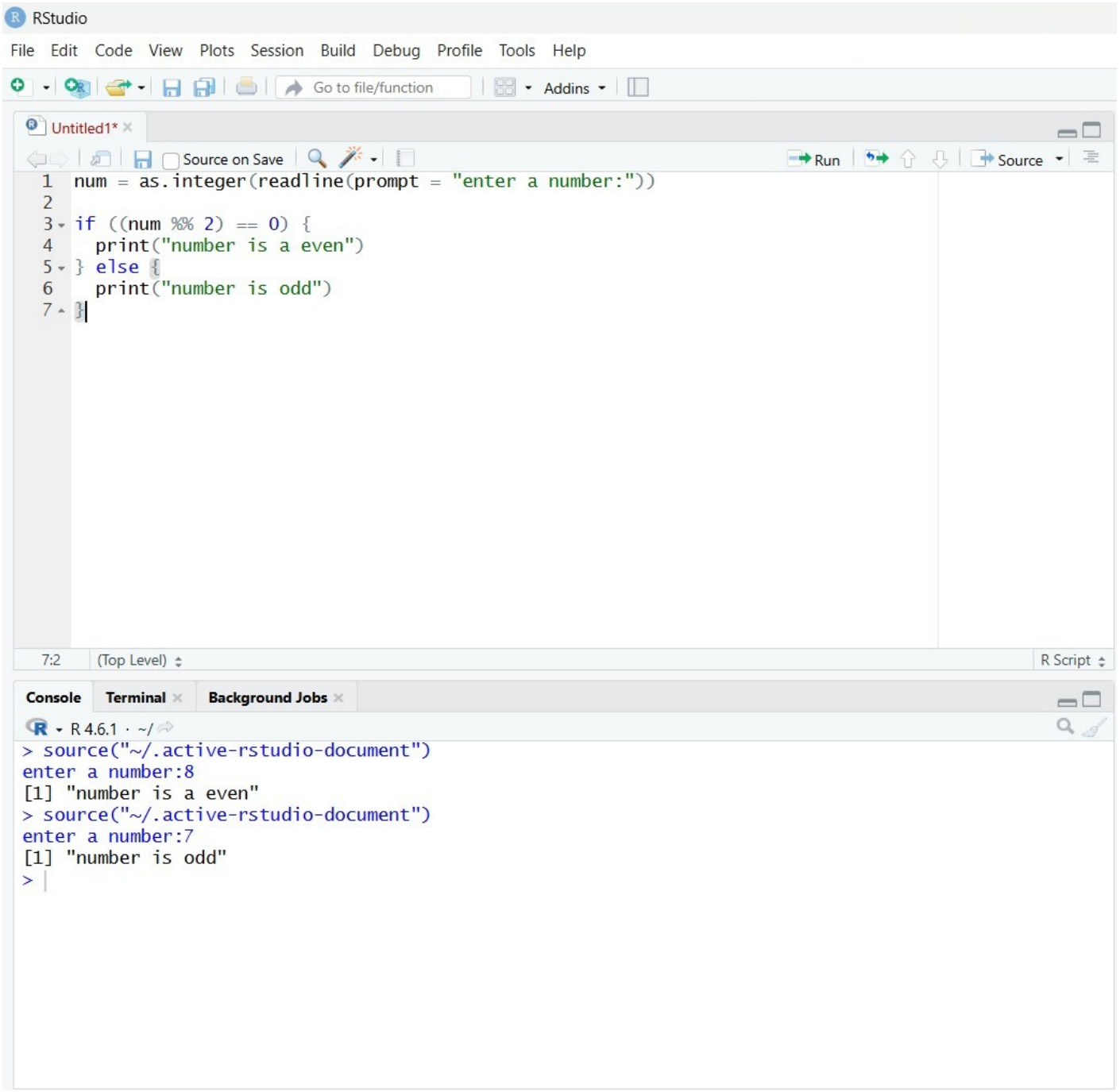
EXPERIMENT-3
MULTIPLICATION



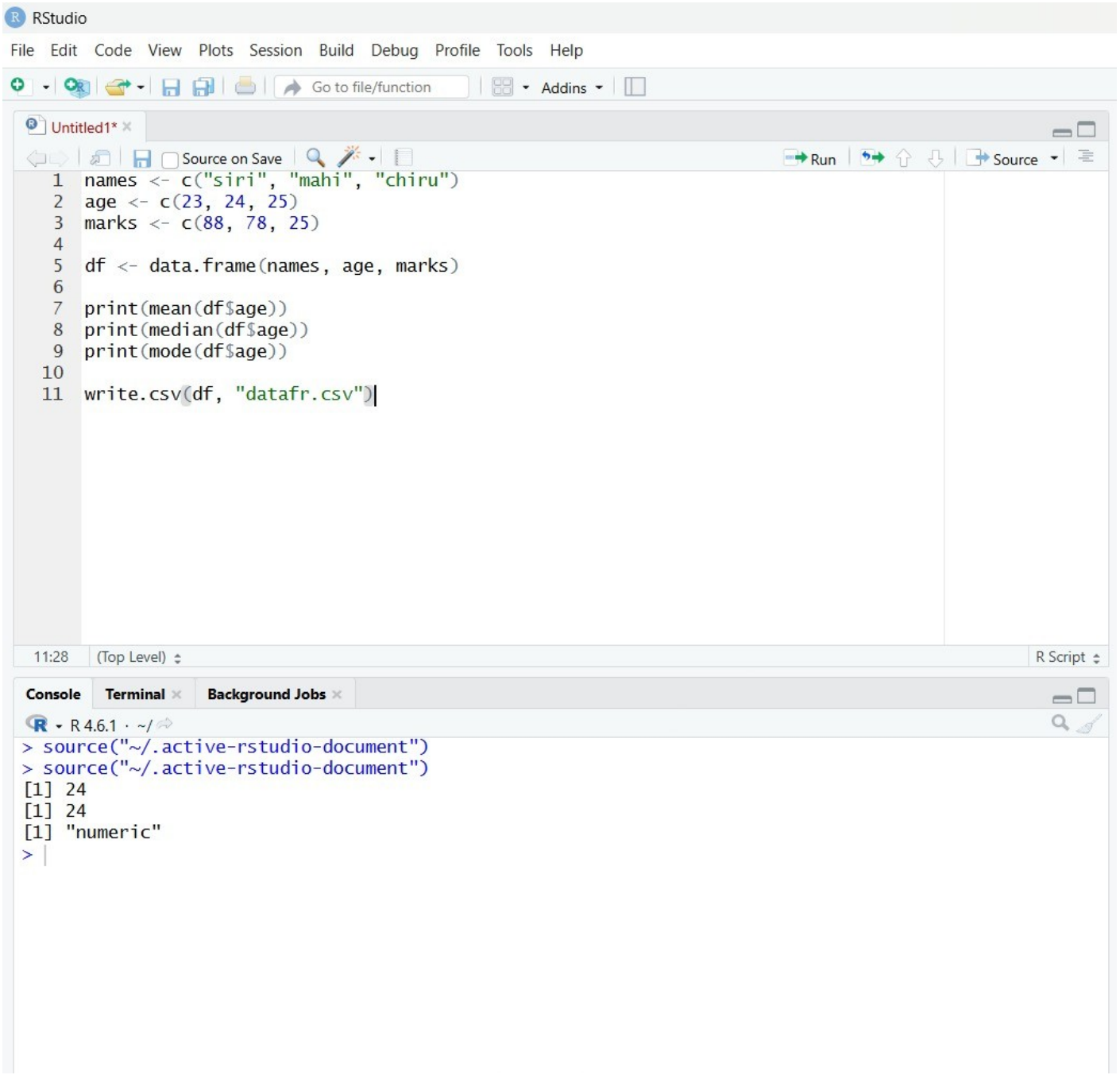
EXPERIMENT-4
DIVISION



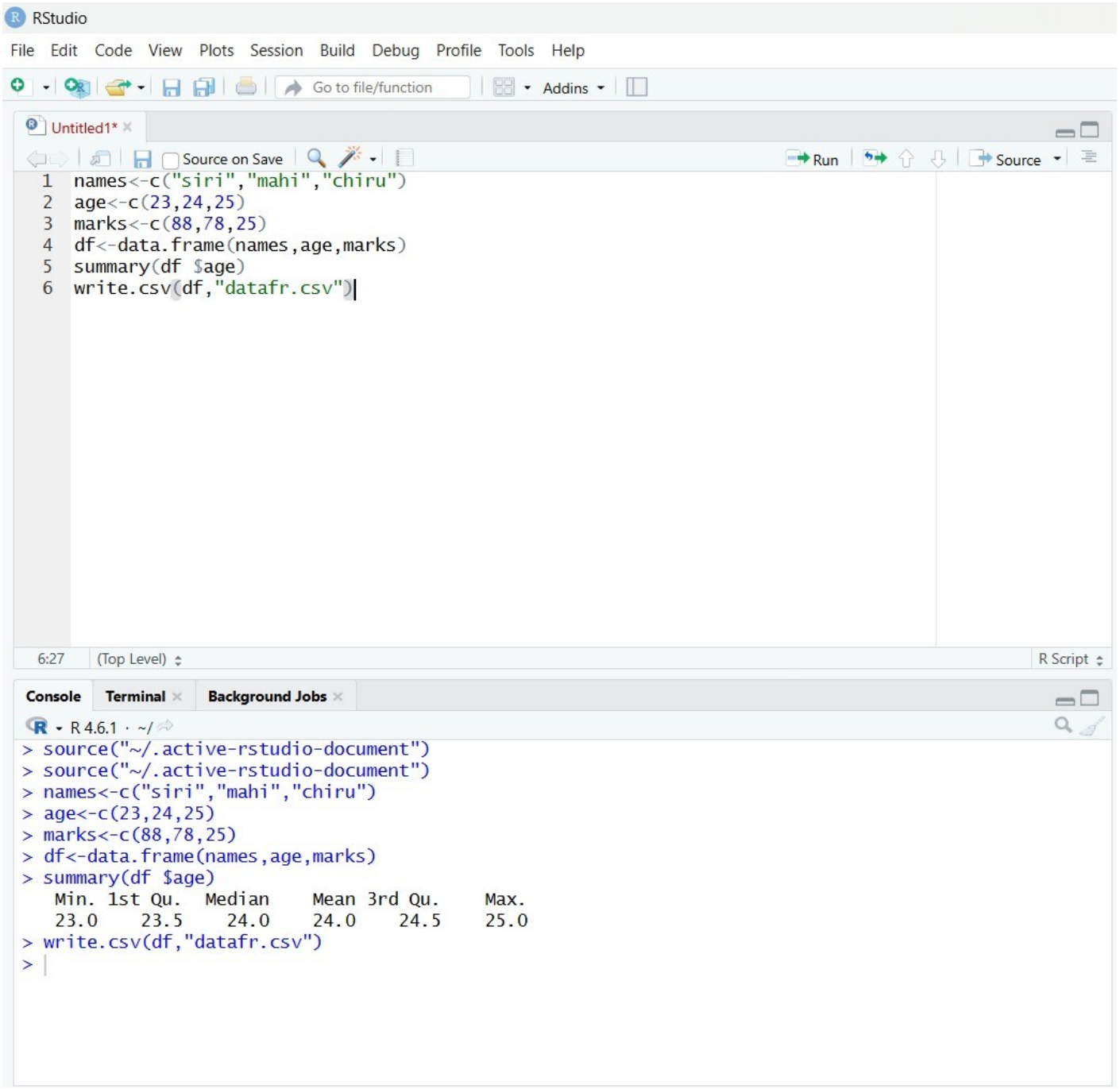
EXPERIMENT-5
ODD OR EVEN



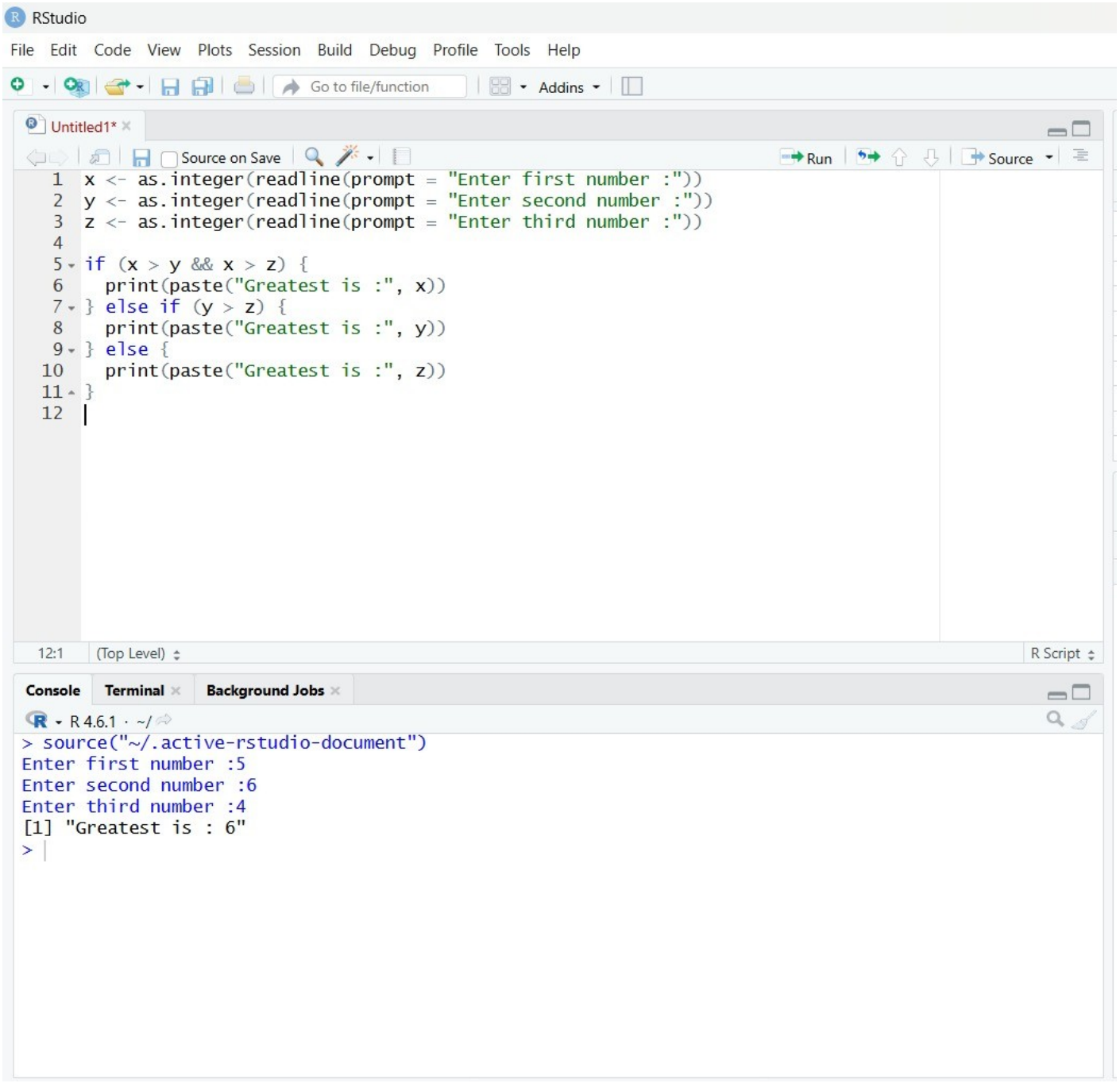
EXPERIMENT-6
MEAN, MEDIAN, MODE



EXPERIMENT-7
SUMMARY



EXPERIMENT-8
GREATEST AMONG THREE NUMBERS



EXPERIMENT-9
IQR

RStudio

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Untitled1*

Source on Save | Run | Source

```
1 names<-c("siri","mahi","chiru")
2 age<-c(23,24,25)
3 marks<-c(88,78,25)
4 df<-data.frame(names,age,marks)
5 IQR(df $age)
6 write.csv(df,"datafr.csv")
```

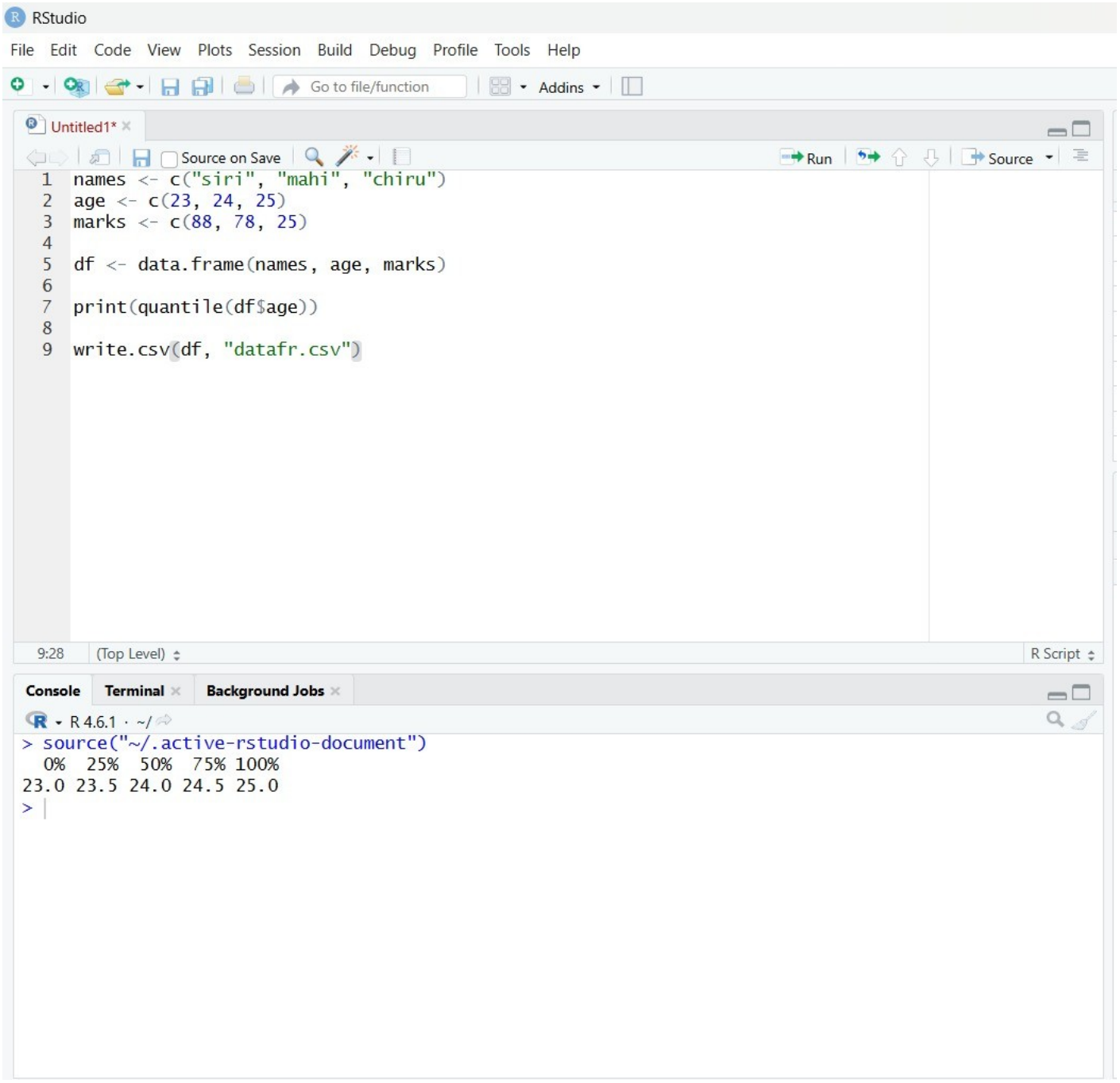
6:27 (Top Level) R Script

Console Terminal Background Jobs

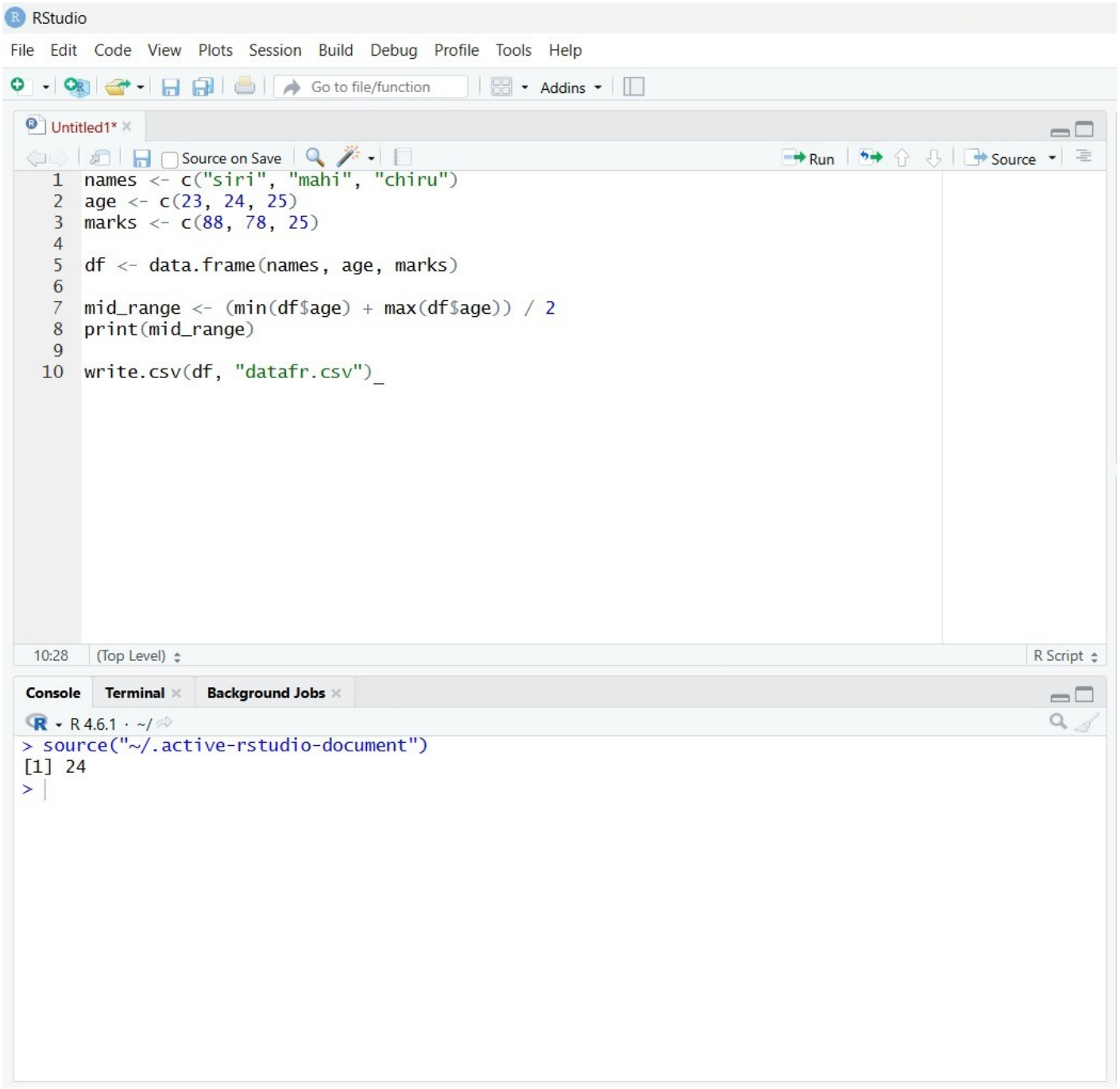
R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
> names<-c("siri","mahi","chiru")
> age<-c(23,24,25)
> marks<-c(88,78,25)
> df<-data.frame(names,age,marks)
> IQR(df $age)
[1] 1
> write.csv(df,"datafr.csv")
>
```

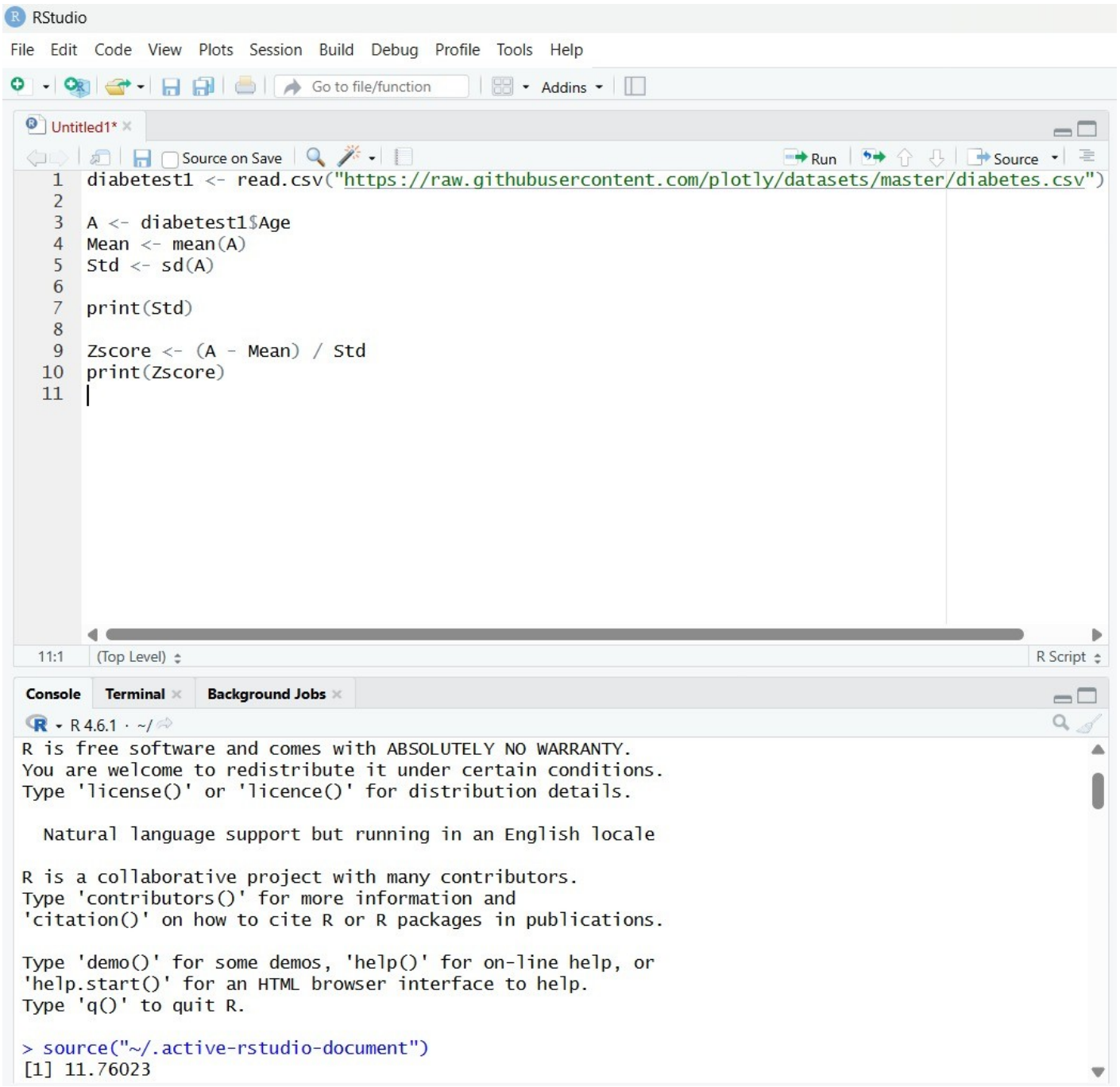
EXPERIMENT-10
QUANTILE



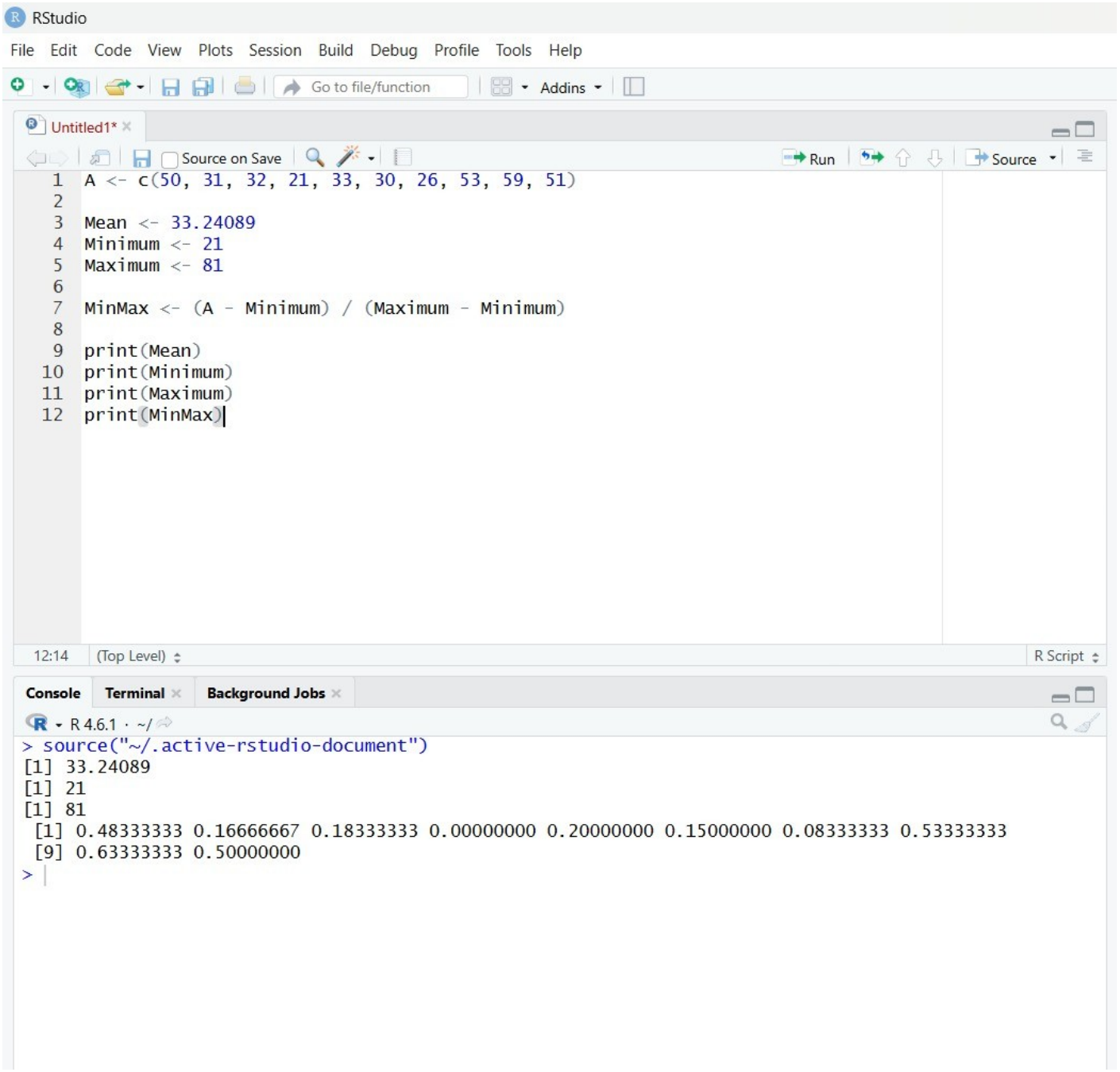
EXPERIMENT-11
MID RANGE



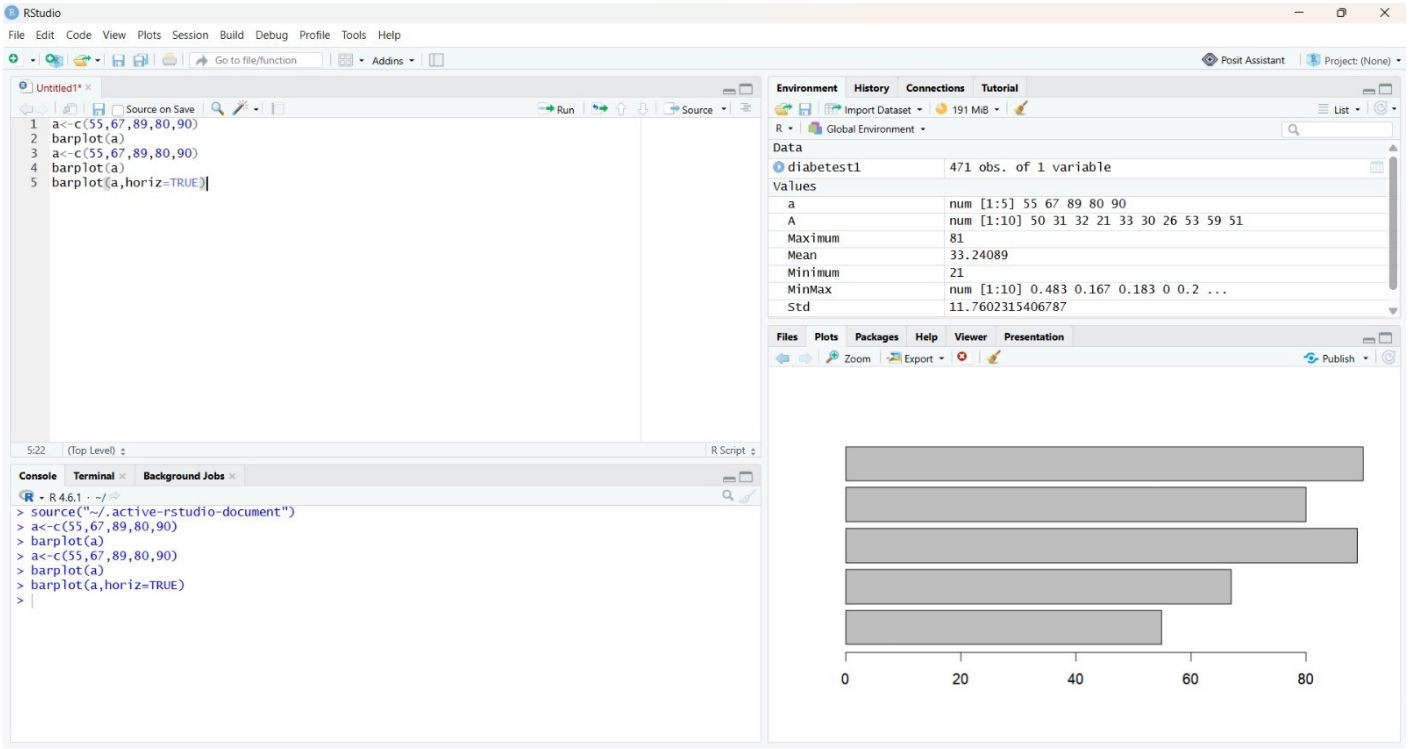
EXPERIMENT-12
Z-SCORE NORMALIZATION



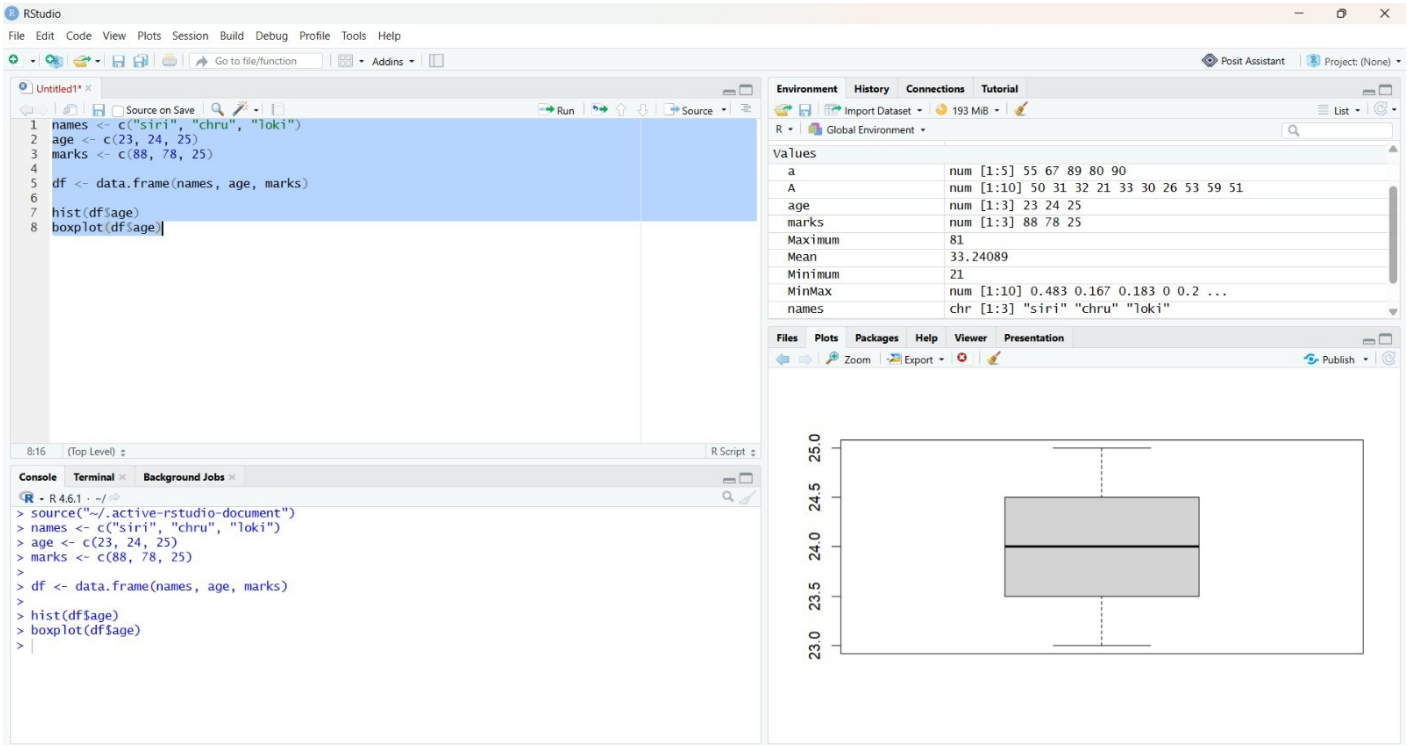
EXPERIMENT-13
MIN, MAX, MEAN, MINMAX



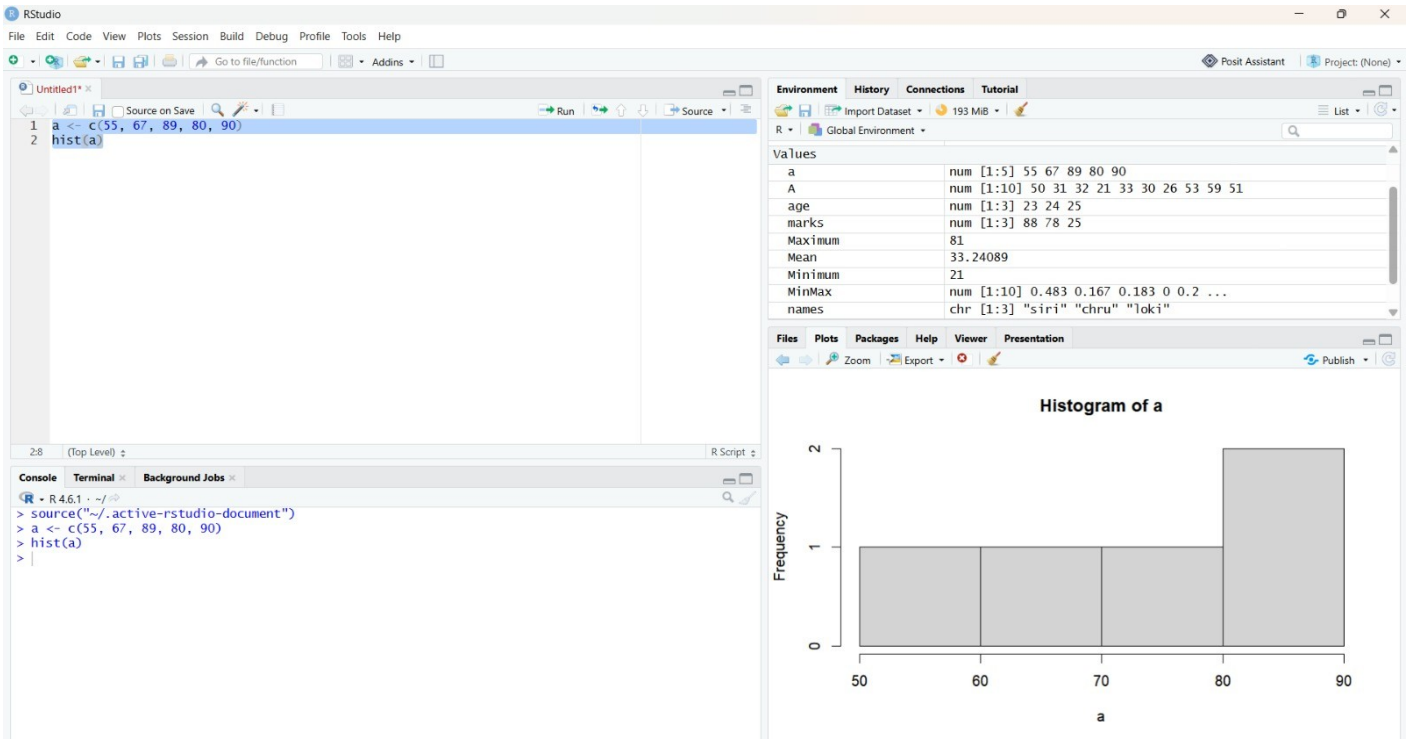
EXPERIMENT-14
BAR PLOT AND HORIZONTAL BAR



EXPERIMENT-15
BOX PLOT



EXPERIMENT-16
HISTOGRAM



EXPERIMENT-17
CORRELATION ANALYSIS

RStudio

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Untitled1*

Source on Save | Run | Source

```
insulin_values <- c(0, 14, 15, 16, 18, 22, 23, 25, 29, 32,
                    36, 37, 38, 40, 41, 42, 43, 44, 45, 46,
                    48, 49, 50, 51)

data <- expand.grid(Age = 21:25, Insulin = insulin_values)

data$Frequency <- 5 + ((data$Age +
                          match(data$Insulin, insulin_values) - 1) %% 4)

Age <- rep(data$Age, data$Frequency)
Insulin <- rep(data$Insulin, data$Frequency)

diabetest1 <- table(Age, Insulin)

print(diabetest1)
print(chisq.test(diabetest1))
```

16:30 (Top Level) R Script

Console Terminal Background Jobs

R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
    Insulin
Age  0 14 15 16 18 22 23 25 29 32 36 37 38 40 41 42 43 44 45 46 48 49 50 51
  21 6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5
  22 7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6
  23 8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7
  24 5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8
  25 6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5  6  7  8  5

        Pearson's Chi-squared test

data:  diabetest1
X-squared = 22.183, df = 92, p-value = 1

> print(chisq.test(diabetest1))
```

EXPERIMENT-18
SCATTER PLOT

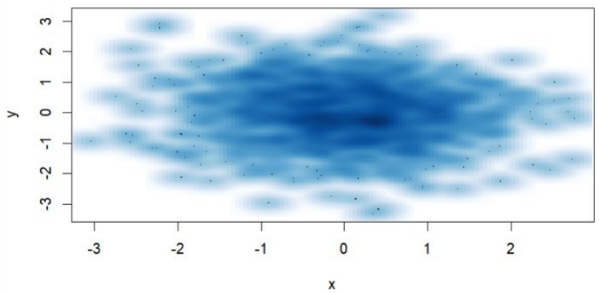
```
set.seed(9)

x <- rnorm(1000)
y <- rnorm(1000)

smoothScatter(y - x)
smoothScatter(x, y)
```

```
> source("~/.active-rstudio-document")
> set.seed(9)
>
> x <- rnorm(1000)
> y <- rnorm(1000)
>
> smoothScatter(y - x)
> smoothScatter(x, y)
> print(chisq.test(diabetest1))
```

insulin_values	num [1:24] 0 14 15 16 18 22 23 25 29 32 ...
marks	num [1:3] 88 78 25
Maximum	81
Mean	33.24089
Minimum	21
MinMax	num [1:10] 0.483 0.167 0.183 0 0.2 ...
names	chr [1:3] "siri" "chru" "loki"
Std	11.760231540678 7
x	num [1:1000] -0.767 -0.816 -0.142 -0.278 0.436 ...
y	num [1:1000] 0.299 -1.127 0.582 0.855 -0.384 ...



EXPERIMENT-19
LINEAR REGRESSION

RStudio

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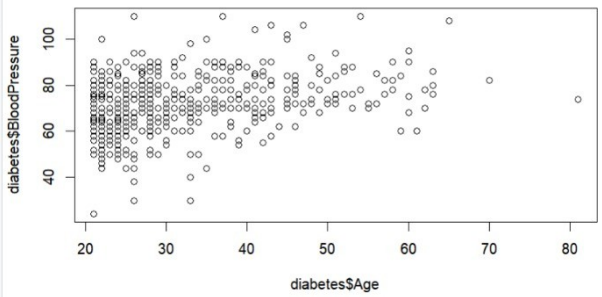
Untitled1*

```
1  library(MASS)
2
3  data(Pima.tr)
4  data(Pima.te)
5
6  diabetes <- rbind(Pima.tr, Pima.te)
7
8  Relation <- lm(bp ~ age, data = diabetes)
9
10 plot(diabetes$age, diabetes$bp,
11      xlab = "diabetes$Age",
12      ylab = "diabetes$BloodPressure")
```

Environment | History | Connections | Tutorial

diabetes	532 obs. of 8 variables
Pima.te	332 obs. of 8 variables
Pima.tr	200 obs. of 8 variables
Relation	List of 12
Values	
a	num [1:5] 55 67 89 80 90
A	num [1:10] 50 31 32 21 33 30 26 53 59 51
age	num [1:3] 23 24 25
Age	int [1:780] 21 21 21 21 21 21 22 22 22 22 ...
diabetest1	'table' int [1:5, 1:24] 6 7 8 5 6 7 8 5 6 7 ...

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Console | Terminal | Background Jobs

R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
> library(MASS)
>
> data(Pima.tr)
> data(Pima.te)
>
> diabetes <- rbind(Pima.tr, Pima.te)
>
> Relation <- lm(bp ~ age, data = diabetes)
>
> plot(diabetes$age, diabetes$bp,
+      xlab = "diabetes$Age",
+      ylab = "diabetes$BloodPressure")
> print(chisq.test(diabetest1))
```

EXPERIMENT-20
MULTIPLE REGRESSION

